

Assignment-7

```
import java.util.*;

class Ring {
    int n;

    int[] process;

    boolean[] active;

    Ring(int n) {
        this.n = n;

        process = new int[n];

        active = new boolean[n];

        for (int i = 0; i < n; i++) {
            process[i] = i + 1; // Process IDs
            active[i] = true; // All processes active initially } }
    }

    void displayProcesses() {
        System.out.println("\nProcess Status:");

        for (int i = 0; i < n; i++) {
            System.out.println("Process " + process[i] +
                " is " + (active[i] ? "Active" : "Down")); } }

    void crashProcess(int id) {
        active[id - 1] = false;

        System.out.println("Process " + id + " is crashed."); }

    void startElection(int initiator) {
        System.out.println("\nElection started by Process " + initiator);

        int maxId = initiator;

        for (int i = 1; i < n; i++) {
            int next = (initiator + i - 1) % n + 1;

            if (active[next - 1]) {
                System.out.println("Process " + next + " participated.");

                if (next > maxId) {
                    maxId = next; } } }
    }
}
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        System.out.println("\nProcess " + maxId + " becomes Coordinator."); }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of processes: ");
        int n = sc.nextInt();

        Ring ring = new Ring(n);
        ring.displayProcesses();

        System.out.print("\nEnter process to crash: ");
        int crash = sc.nextInt();
        ring.crashProcess(crash);
        ring.displayProcesses();

        System.out.print("\nEnter process to start election: ");
        int initiator = sc.nextInt();
        ring.startElection(initiator); } }

```

Bully.java

```

import java.util.*;

class Bully {
    int n;
    int[] process;
    boolean[] active;
    int coordinator;

    Bully(int n) {
        this.n = n;
        process = new int[n];
        active = new boolean[n];
        for (int i = 0; i < n; i++) {
            process[i] = i + 1;
            active[i] = true; }
        coordinator = n; // highest ID initially coordinator }

    void displayProcesses() {
        System.out.println("\nProcess Status:");
    }
}

```

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for (int i = 0; i < n; i++) {
    System.out.println("Process " + process[i] +
        " is " + (active[i] ? "Active" : "Down"));    }
    System.out.println("Current Coordinator: " + coordinator); }
void crashProcess(int id) {
    active[id - 1] = false;
    System.out.println("Process " + id + " crashed."); }
void startElection(int initiator) {
    System.out.println("\nElection started by Process " + initiator);
    boolean higherActive = false;
    for (int i = initiator; i < n; i++) {
        if (active[i]) {
            System.out.println("Process " + (i + 1) +
                " responds OK.");
            higherActive = true;        }    }
    if (!higherActive) {
        coordinator = initiator;
        System.out.println("Process " + coordinator +
            " becomes new Coordinator.");
    } else {
        System.out.println("Higher process will conduct election.");    } }
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of processes: ");
    int n = sc.nextInt();
    Bully bully = new Bully(n);
    bully.displayProcesses();
    System.out.print("\nEnter process to crash: ");
    int crash = sc.nextInt();
    bully.crashProcess(crash);
    bully.displayProcesses();
}

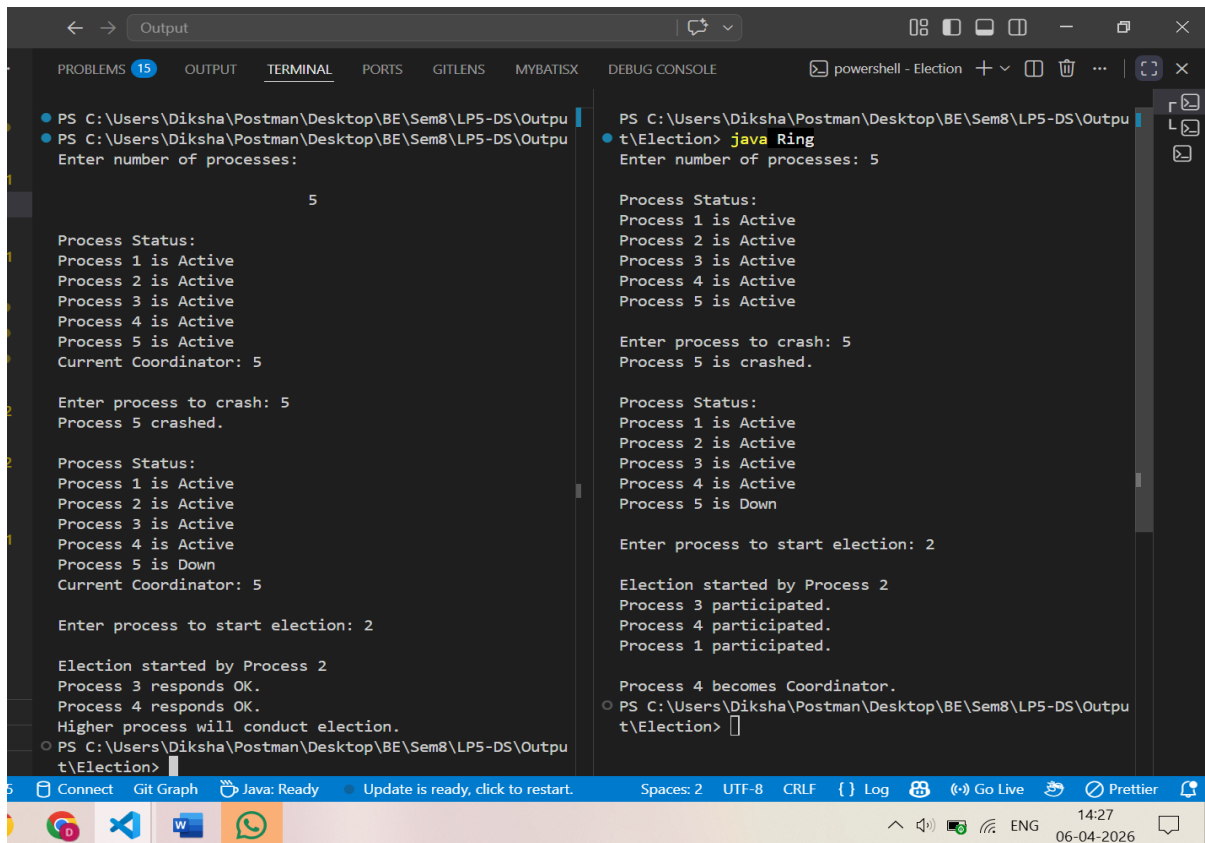
```

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System.out.print("\nEnter process to start election: ");
```

```
int initiator = sc.nextInt();
```

```
bully.startElection(initiator); }}
```

Output:-



```
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output
Enter number of processes:
5
Process Status:
Process 1 is Active
Process 2 is Active
Process 3 is Active
Process 4 is Active
Process 5 is Active
Current Coordinator: 5
Enter process to crash: 5
Process 5 crashed.
Process Status:
Process 1 is Active
Process 2 is Active
Process 3 is Active
Process 4 is Active
Process 5 is Down
Current Coordinator: 5
Enter process to start election: 2
Election started by Process 2
Process 3 responds OK.
Process 4 responds OK.
Higher process will conduct election.
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output
t\Election>
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output
t\Election> java Ring
Enter number of processes: 5
Process Status:
Process 1 is Active
Process 2 is Active
Process 3 is Active
Process 4 is Active
Process 5 is Active
Enter process to crash: 5
Process 5 is crashed.
Process Status:
Process 1 is Active
Process 2 is Active
Process 3 is Active
Process 4 is Active
Process 5 is Down
Enter process to start election: 2
Election started by Process 2
Process 3 participated.
Process 4 participated.
Process 1 participated.
Process 4 becomes Coordinator.
PS C:\Users\Diksha\Postman\Desktop\BE\Sem8\LP5-DS\Output
t\Election>
```